COI1 (O04197 / At2g39940, *Arabidopsis thaliana*) — Function-Assignment Hypothesis Review

Hypothesis (seed): *COI1 has a molecular role in root defense-gene repression independent of canonical JA-Ile/JAZ perception.* **Focus type:** function_assignment · **Slug:** ja-independent-root-repression **Primary reference under audit:**

P 34145662

(Ulrich, Schmitz, Thurow, Gatz, *Plant J.* 2021)

Executive Judgment

Verdict: PARTIALLY SUPPORTED — supported at the process/genetic-requirement level, UNRESOLVED at the molecular-function level.

The published data (P 34145662, extended by P 39945499) genetically establish a **phenotype**: COI1 is *required in roots to keep a defined set of defense/immunity genes constitutively repressed*, and this requirement is **independent of the JA-Ile ligand** (the JA-Ile–deficient *aos* mutant does not de-repress) **and independent of canonical JAZ perception** (a JAZ-interaction–compromised, JA-signaling–dead COI1_AA variant still represses). That is a legitimate, well-controlled genetic dissociation.

However, the hypothesis as worded claims a **"molecular role"** for COI1 in repression. The evidence does **not** reach that bar:

1. **No molecular activity, substrate, partner, or DNA/chromatin association** is assigned to COI1 for this repression. The authors themselves label it a *"potential moonlighting function"* — explicitly provisional.
2. **SCF-dependence is not excluded.** COI1_AA retains its F-box (16–57) and therefore still assembles into SCF^COI1. "Independent of canonical *perception*" (JA-Ile + JAZ) is demonstrated; **independence from the SCF/ubiquitin machinery is untested.** The mechanism could still be SCF^COI1-mediated ubiquitination of a *non-JAZ* substrate.
3. The phenotype is a **loss-of-function de-repression**; direction of causality (COI1 directly represses vs. COI1 sustains a downstream repressor / degrades an activator) is not resolved.

Bottom line for the curator: The JA-Ile/JAZ-independence claim is real and should be captured, but it supports a **Biological-Process negative-regulation annotation with an explicit "JA-Ile/JAZ-independent" note**, *not* a new Molecular-Function assignment. Do not annotate a molecular repression activity to COI1 from these data.

Independent reproduction (this review, Iteration 2). I reanalyzed GEO **GSE282297** (P 39945499) mock root raw counts (27,628 genes; coi1 n=4, matched WT n=3, aos n=4, matched WT n=4; CPM/log2, Welch t-test, BH-FDR<0.05, |log2FC|>1). Genotype QC passed (AOS transcript 5.66→2.02 log2CPM in *aos*, ~13×; COI1 6.38→5.14 in coi1). **coi1 vs WT = 24 UP / 2 DOWN; aos (JA-Ile-null) vs WT = 0 UP / 0 DOWN; coi1 vs aos = 132 UP / 1 DOWN.** The near-total up/down asymmetry and, critically, the **null result for *aos* beside strong coi1 de-repression**, independently reproduce the JA-Ile-independence discriminator. (Fewer genes than the paper's 316 reflect the stricter simple test and low replication, not a directional discrepancy.) Provenance: `coi_derepression_barplot.png`, `coi_vs_aos_upgenes.csv`.

Evidence Matrix

Citation	Evidence type	Stance	Claim tested	Key finding	Context
P 34145662 (Plant J 2021)	Mutant transcriptomics + complementation	**Supports** (JA-Ile & JAZ independence)	Is COI1-dependent root repression independent of JA-Ile and JAZ?	coi1-null de-represses genes; JA-Ile–null *aos* does **not** (→ JA-Ile independent). COI1_AA (JAZ-interaction compromised; weak VSP2, sterile) **still represses** target genes (→ JAZ independent). Called "potential moonlighting function."	*A. thaliana* roots; whole-root transcriptome; wounding/ VSP2 as JA control
P 39945499 (J Exp Bot 2025, same group)	Mutant transcriptomics + functional genetics	**Supports / qualifies**	Reproducibility & biological meaning of coi1 de-repression	316 immunity genes constitutively higher in coi1 vs *aos*/WT; knocking down a subset partially reduces coi1 tolerance; repression **lifted/ overridden upon *V. longisporum* infection**.	*A. thaliana* roots; Verticillium pathosystem
P 20927106 (Nature	Structural / biochemical (direct)	**Competing / context**	What is COI1's *canonical*	COI1 + JAZ + JA-Ile + inositol-pentakisphosphate form the co-	In vitro / structural;

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2010, Sheard et al.)			molecular function?	receptor; COI1 is the F-box substrate-receptor driving JA-Ile–dependent JAZ ubiquitylation/ degradation.	Arabidopsis proteins
P 16732289 (PNAS 2006, Consonni et al.)	Genetic epistasis (leaf)	**Separate claim**	Does *mlo* powdery-mildew resistance need JA (COI1)?	*mlo* resistance is **JA/SA/ethylene-independent**; needs PEN1 syntaxin, PEN2 glycosyl hydrolase, PEN3 ABC transporter.	*A. thaliana* leaves/ epidermis; *Blumeria/ Erysiphe* powdery mildew
GSE282297 reanalysis (this review; data = P 39945499)	Independent computational reproduction	**Supports** (JA-Ile independence)	Does JA-Ile loss (aos) phenocopy coi1 de-repression?	coi1 vs WT 24↑/ 2↓; **aos vs WT 0↑/ 0↓**; coi1 vs aos 132↑/1↓. Genotype QC passed. Top coi1-up genes defense/ secretion-related.	A. thaliana mock roots; Welch t, BH-FDR

Citation	Evidence type	Stance	Claim tested	Key finding	Context
UniProt O04197 (database)	Sequence/domain + curated GO	Orientation	Protein identity & existing MF/BP	F-box (16–57) + 18 LRR; SCF ubiquitin-ligase complex (GO:0019005, IDA); already carries **GO:0031348 negative regulation of defense response (IMP:TAIR)** and GO:0050832 defense response to fungus (IMP:TAIR).	Reference proteome

GO Curation Implications (leads — require curator verification)

- **Molecular Function:** *No new MF term is supported.* COI1's demonstrated MF remains F-box / jasmonate co-receptor / SCF substrate-recognition subunit (well grounded by P 20927106). The repression data are genetic, not biochemical — **do not add an MF "repressor/transcriptional" activity**, and **do not fall back to "protein binding."**
- **Biological Process (primary lead):** The data support / reinforce **GO:0031348 "negative regulation of defense response"** (and, more specifically, negative regulation of defense/immunity gene expression in root), with **IMP** evidence from *coi1* and the COI1_AA complementation. Recommend **retaining** GO:0031348 and, if the review is considering removing or narrowing it, the primary literature *supports keeping it* — with an explicit annotation comment that this repression is **JA-Ile- and JAZ-independent** (non-canonical), tissue = root.

- **Cellular Component:** No change warranted from these data (nucleus/cytoplasm/SCF as already annotated).
- **Do NOT** let

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  (leaf, powdery mildew, JA-independent *mlo* resistance) validate or extend any COI1 defense annotation; it is a separate, largely *absence-of-role* context.

Suggested qualifier language for the curator: "COI1 is genetically required to repress a set of root defense/immunity genes independently of JA-Ile and JAZ (

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;

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); the underlying molecular mechanism, including possible SCF^COI1-dependence, is unresolved."

GO Decision Table (leads — require curator verification)

GO term	Aspect	Current status	Evidence basis	Recommended action
GO:0031348 negative regulation of defense response	BP	Present (IMP:TAIR)	coi1-null de-represses root defense genes; JA-Ile-independent (P 34145662 ; P 39945499 ; reproduced here)	**Retain**; add comment "root; JA-Ile/JAZ-independent; mechanism unresolved"
(new, more specific) negative regulation of defense/immune-system-process gene expression	BP	Absent	Same as above; effect is on constitutive transcript levels	**Optional add** as narrower child, IMP, with same qualifier — curator's discretion
(proposed) any new MF "transcriptional repressor / repression activity"	MF	Absent	No biochemical activity, substrate, DNA/chromatin binding shown	**Do NOT add**; genetic requirement only
GO:0019005 SCF ubiquitin ligase complex / F-box co-receptor (canonical MF/CC)	MF/CC	Present (IDA)	P 20927106 structure	**Retain unchanged**; the non-canonical repression does not modify canonical MF
"protein binding" (GO:0005515) as summary of this function	MF	—	Not informative for this claim	**Avoid** as a recommendation

Mechanistic Scope

- **Immediate molecular function being tested:** whether COI1 protein directly effects gene repression via a non-canonical activity.
- **What is directly shown:** a *requirement* (loss-of-function de-repression) — a genetic/transcriptomic dependency, not a biochemical activity.
- **Downstream / not-COI1-activity:** the de-repressed genes, the secreted-defense-compound hypothesis, Verticillium tolerance, and infection-triggered lifting of repression are **downstream phenotypes**, not COI1 molecular functions.
- **Canonical function (separate):** JA-Ile–dependent SCF^COI1-driven JAZ ubiquitylation/degradation (P 20927106) — the pathway the hypothesis claims to bypass.

Conflicts and Alternatives

1. **SCF-dependent alternative (strongest competitor):** Because COI1_AA keeps its F-box, the "non-canonical" repression may still be **SCF^COI1-mediated ubiquitination of an unknown non-JAZ substrate** — this would make it *perception-independent but machinery-dependent*, not a truly ligase-independent "moonlighting" activity. Untested.
2. **Residual-JAZ alternative:** COI1_AA is *compromised*, not certified null, for JAZ interaction; a low-level, ligand-independent JAZ interaction cannot be fully excluded by the abstract-level evidence.
3. **Indirect/basal-signaling consequence:** de-repression in a null mutant could reflect a developmental or basal-signaling secondary effect rather than direct COI1 repressor action. **However**, my GSE282297 reanalysis argues against a *JA-Ile-dependent* basal-signaling explanation: *aos* (which also lacks basal JA-Ile signaling) shows 0 de-repressed genes, so the de-repression cannot be attributed to loss of basal JA-Ile flux. A JA-Ile-*independent* indirect/developmental route remains possible and is not excluded.
4. **Paralog confusion:** low risk — O04197 is the single-copy Arabidopsis JA receptor; F-box + 18-LRR architecture confirmed. Distinguish from JAZ family and from COI1 orthologs in other species.

5. **Context conflation:**

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   (leaf/powdery-mildew) must be kept distinct from the root/Verticillium repression claim.

Knowledge Gaps (what would change the verdict)

Gap	Checked?	Why it matters	Resolving evidence
SCF^COI1 dependence of the repression	Not addressed in cited papers	Determines whether "independent of canonical perception" also means "independent of E3-ligase activity"; decisive for any MF claim	Complement coi1 with an F-box–disrupted / Cullin-binding–dead COI1; test if repression is lost
Direct molecular partner/substrate of COI1 for repression	None identified (2021 & 2025)	Required to assign any *molecular role* rather than a genetic requirement	IP-MS / proximity labeling of COI1 (or COI1_AA) in roots; identify a repression-relevant, non-JAZ interactor/substrate
Direct vs. indirect repression	Not resolved	"Molecular role in repression" needs directness	ChIP/CUT&RUN for COI1 near target loci; rescue kinetics
Certified JAZ-null status of COI1_AA	Partial (functional JA-dead)	Underpins the JAZ-independence claim	In vitro JAZ-binding of COI1_AA; higher-order jaz mutant epistasis
Root-specificity / tissue scope	Root-only tested	Prevents over-generalization to leaf annotations	Parallel leaf transcriptomics of coi1 vs aos

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Discriminating Tests (most efficient)

1. **F-box–dead / Cullin-binding–dead COI1 complementation** of coi1 → if repression is lost, the function is SCF-dependent (refines "independent of perception" but not of machinery); if retained, supports a truly ligase-independent moonlighting activity. *(Single most decisive experiment.)*
2. **COI1 interactome in roots (IP-MS / TurboID)** using COI1 vs COI1_AA → seek a non-JAZ partner/substrate that could confer a molecular repression role.
3. **Higher-order *jaz* / *myc2,3,4* epistasis** on the coi1-repressed gene set → test whether any residual JAZ/MYC route contributes.
4. **ChIP/CUT&RUN for COI1** at target loci → directness of repression.
5. **Re-analysis of the coi1-vs-aos-vs-WT root RNA-seq** (P 34145662 / GEO) to independently reproduce the de-repressed gene set and check for SCF/proteasome-pathway signatures.

Curation Leads (require curator verification)

- **Candidate references to attach:**
- P 34145662 — verify snippet: *"genes affected by COI1 but not by JA-Ile were still strongly repressed by COI1_AA"* and *"COI1 has a potential moonlighting function that serves to repress gene expression in a JA-Ile- and JAZ-independent manner."*
- P 39945499 — verify snippet: *"316 immunity-related genes that were constitutively higher expressed in coi1 as compared to the susceptible genotypes."*
- **Candidate GO action: Retain** GO:0031348 (negative regulation of defense response, IMP), add a curator note "root, JA-Ile/JAZ-independent (P 34145662 ;

)."

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Consider a more specific BP child (negative regulation of defense/immune gene expression) if the ontology permits. **Do not add a Molecular-Function repression term.**

- **Action change on the seed hypothesis:** downgrade from "molecular role in repression" to "genetically required for JA-Ile/JAZ-independent repression of root defense genes (mechanism unresolved; SCF-dependence untested)."
- **Suggested curator questions:** (1) Is SCF-dependence known/tested anywhere? (2) Should

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  be de-linked from this root claim? (3) Is there a GEO accession for the 2021 RNA-seq to reproduce the gene set?
- **Suggested experiments:** F-box–dead complementation; COI1 root interactome — see Discriminating Tests.

Limitations of this Review

- Assessment is abstract- and database-level for the primary papers (full texts not machine-read here); the full 2021 abstract was recovered via EuropePMC after `search_pubmed` returned a truncated version. Snippets above should be re-verified against the published text.
- Local `*-bioinformatics` analyses and repository reviews were intentionally not consulted, per instructions.
- No claim of catalytic/regulatory molecular function is inferred from fold, interaction, or docking.

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